

Study of Ambient Air Methane Concentration at Maitri, Antarctica

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ABSTRACT

To study the ambient air concentration of methane at background site, measurement on hourly basis were carried out during the 23rd IASE in the year 2004 at Maitri, Antarctica (70° 45' S, 11° 45' E, 140 masl). Global warming studies require reliable estimates of true baseline values for greenhouse gases, which is only possible at some pristine location like Antarctica. Measurement of CH₄ made for a period of 22 months showed that the day-to-day variation was small except the variation of few ppb. No diurnal variation has been observed. About 83% of CH₄ values have been observed in the range of 1.685 ppmv to 1.715 ppmv during the entire period of measurement from February 2003 to December 2004.

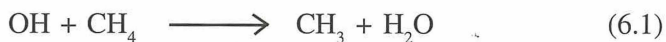
The ambient air surface concentration of CH₄ during the period of observation was found to be nearly stable. Annual mean concentration of 1.699 ppmv for CH₄ has been observed in 2003 with standard deviation of 13 ppbv and 1.701 ppmv in 2004 with standard deviation of 6 ppbv. Mean CH₄ concentration for 22 month period has been observed at 1.7 ppmv with standard deviation of 9 ppbv. The observed stable CH₄ concentration indicates a decrease in emissions, which may have forced the global methane budget towards a steady state.

Keywords: Methane; Antarctica; Global warming; baseline values

INTRODUCTION

The distribution of greenhouse gases (GHGs) in the atmosphere, particularly those due to human and biosphere activities, is quite heterogeneous (Mukhopadhyay et al, 2002). Increase in the atmospheric concentration of GHGs such as carbon dioxide, methane, nitrous oxide,

halo-carbons and tropospheric ozone since the industrial period (1750) has been identified as one of the major causes of the warming of the earth's surface (Hansen et al, 1998; Lacis et al, 1981; Wang et al, 1976). CO₂ has the largest radiative effect. Now it has been recognized that the global warming due to infrared active gases like O₃, CFCs, CH₄, N₂O, etc. is collectively as much as that due to CO₂ alone (Jain et al, 2000), where as it was only due to CO₂ during the pre-industrial era upto 1950s. Next to carbon dioxide, methane is the most abundant and important organic trace gas in the atmosphere. It absorbs infrared radiation (Lelieveld et al, 1998) and is also involved in tropospheric and stratospheric photochemical reactions (Lelieveld and Crutzen, 1992). The major natural sources of Methane are natural wetlands, termites and oceans and anthropogenic sources are rice paddies, energy landfills, ruminants, waste treatment and biomass burning (IPCC, 2001; Schimel et al, 1996). The main sink of atmospheric methane is the reaction with OH radical (Singh et al, 1995):



The reaction is rather slow and, as a result, the atmospheric lifetime of methane is long. Recent process model study by Ridgwell (1999) suggested that soil sinks CH₄ of approximately 30Tg/yr. Similarly, stratospheric loss is another minor sink of methane (Hein et al, 1997). Methane has strong absorption band at 3.3 μm and 7.66 μm and trap part of the thermal radiation from the earth's surface (Wang et al, 1976). Lelieveld et al (1998) and Jain et al (2000) estimated that the current direct radiative forcing of 0.48 Wm⁻² from CH₄ is 20% of the total from all of the long-lived and globally mixed greenhouse gases. Based on the reaction rate coefficient between CH₄ and OH, and the estimated tropospheric distribution of OH, Prinn et al (1995) derived a mean atmospheric lifetime for CH₄ of 8.9 years. In the stratosphere, methane is a source of water vapor and hydrogen and controls the abundance of Cl atoms converting them to HCl as,



Although some of the hydrogen is produced by CH₄, most of the CH₄ molecule transported into stratosphere produces water vapor by oxidation. Craig and Chou (1982) observed that, in the distant past, variations in natural sources of methane were responsible for trends in atmospheric methane levels recorded in ice cores. On the basis of analysis of ancient air trapped

in polar ice cores, Rasmussen and Khalil (1984) reported the increase in growth rate of CH_4 concentration during the 19th century. Measurements of CH_4 in air extracted from ice cores and furn showed that the atmospheric abundance of CH_4 has increased by about a factor of 2.5 since the pre-industrial era (Etheridge, 1998). On the basis of measurements from 1992 to 1998, Dlugokencky (2001) derived a global tropospheric CH_4 growth rate of about 4.9 ppb/yr. A downward trend was observed in the surface concentrations of methane in the northern and southern hemispheres between 1987 and 1992 and even larger downward trends were observed (Dlugokencky, 2001) for the period 1992–93. These trends have again reversed to the previous upward trend of almost 1% for methane. Dlugokencky et al (2001) reported that the global methane increase averaged 3.9 ppb per year during the three-year period 1995–97, but it was 12.7 ppb in 1998. In 1999 the increase was 2.6 ppb, suggesting that the large increase in 1998 was temporary (Dlugokencky, 2001). The large increase in global methane abundance during 1998 corresponds to an increase in methane emissions of about 5% of the total emissions. Dlugokencky et al (2003) noted that the globally averaged atmospheric methane abundance was constant at ~1751 ppb from 1999 through 2002, which suggests that during this four-year period the global methane budget has been at steady state. Dlugokencky et al (1997) reported methane measurement by gas chromatography with flame ionization detector. There is still considerable uncertainty associated with global methane budget due to the unavailability of methane distributions for many areas. Also, even if the main climatic effects of CH_4 are not local, it is important to measure baseline concentration of CH_4 at unpolluted sites like Antarctica, where human activities are at a bare minimum. The changing behavior of the atmospheric trace gases, which are very fundamental to climate change, can be understood with the coverage of global measurements.

In view of the above NPL has taken a lead in measurement of methane during the 22nd (2003) and 23rd (2004) Indian Scientific Expedition at Maitri, Antarctica respectively. The continuous measurements will sequentially provide valuable data for modeling studies. Studies at Maitri station were aimed towards the estimation of baseline concentration of CH_4 , the annual trend and day-to-day variation in the near-surface air in Antarctica.

EXPERIMENTAL SETUP AT SITE

The instrumentation system established at Maitri, Antarctica for the online measurement of surface air concentration of CO_2 and CH_4 using gas chromatograph is as shown in Figure 1. The complete system incorporates

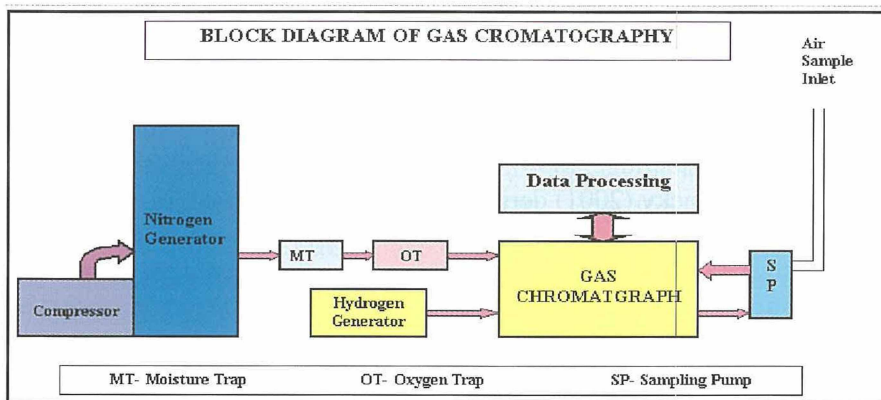


Fig. 1: Block Diagram of Gas Chromatograph system established at Maitri in 2002.

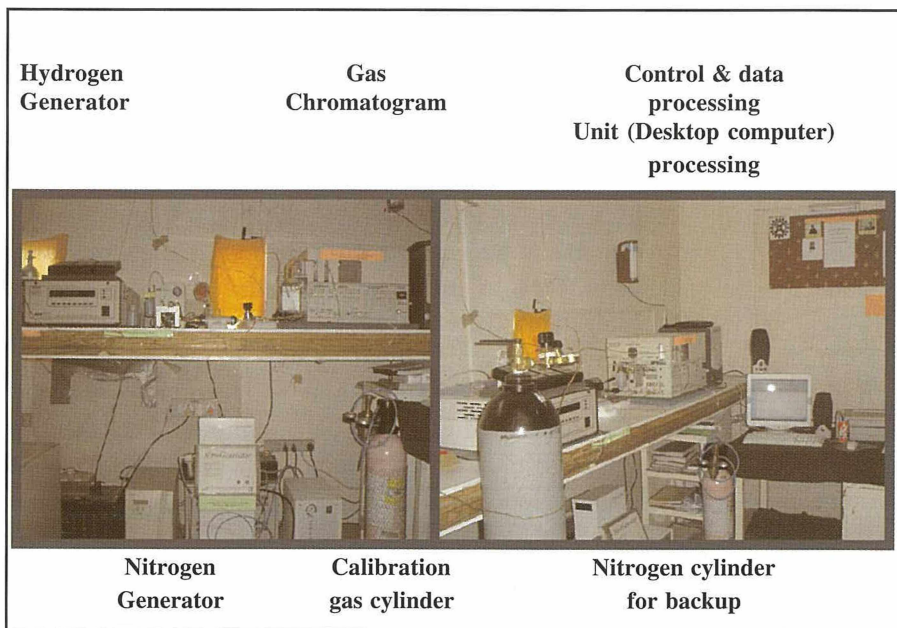


Fig. 2: Photograph of Gas chromatogram system installed at Maitri, Antarctica.

a Nitrogen Generator, Hydrogen Generator, computer controlled Sampling Pump, Air Compressor, Gas Chromatograph and Data Processing Unit. The instrumentation system has an independent nitrogen generator producing ultra high pure nitrogen as carrier gas at site that transports the sample components through the column to the detector. The system also has

separate Hydrogen Generator and inbuilt Air Compressor which is needed as a fuel to produce the flame for FID. The sample from ambient air is injected into the column by an online sampling pump for known programmed interval of time. Molecular sieve 13X column has been used to determine surface concentration of CH_4 in Antarctic air. The detector system incorporates FID, amplifier and processing unit. FID has high sensitivity to virtually all organic compounds, no response to water and CO_2 and good linearity over a wide sample concentration range.

The main air inlet was installed on top of the sidewall of the observatory, 3 m above the ground. The copper tube between the inlet and the sampling pump was around 3 m long with an inner diameter of 2 mm. The inlet was protected against drifting snow with a specially designed tumbler and no experiment was performed during the heavy snowdrift and Antarctica blizzard. To achieve optimum separation, good accuracy and precision, good regulation of pressure and flow rate of mobile phase and temperature programming of the column is done using computer controlled software.

The data was collected on an hourly basis for approximately 12 to 14 hours a day for six days a week and round the clock once a week. The instrument was regularly calibrated once a week with Linde Standard Gas. Conditioning of Moisture trap in nitrogen line and hydrogen line was done twice a month and column conditioning and other maintenance tasks were carried out once a month on regular basis. Figure 2 shows the photograph of the GC system installed in the Environmental Laboratory at Maitri, Antarctica. A typical chromatogram obtained as on 08-02-2003 is shown in Figure 3.

I) OBSERVATION AND RESULTS

Day-to-day variability:

Variation of daily mean CH_4 concentration for the observational period of February 2003 to December 2004 is depicted in Figure 4.

Hourly sampled data were averaged and mean CH_4 concentration has been calculated for each day. Due to baseline drift in the chromatogram (due to malfunctioning of the instrument) and runaway from calibration, incorrect CH_4 values were carefully filtered out from the dataset. The data gaps in Figure 4 are basically due to snow drift during Antarctic blizzards when GC had to be shut down because of snow ingress. Secondly due to

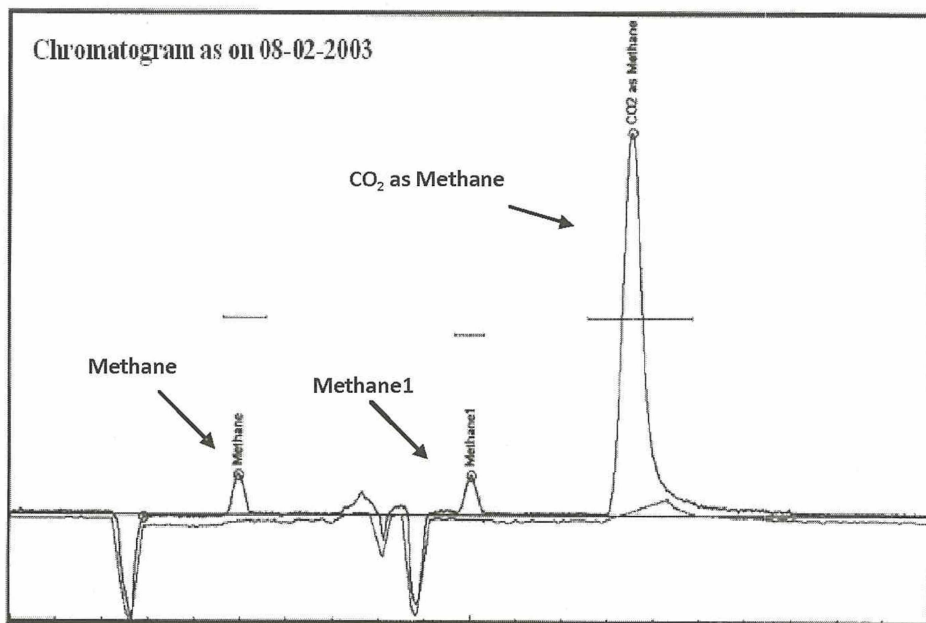


Fig. 3: Typical chromatogram of air samples obtained on 8 February 2003 at Maitri, Antarctica

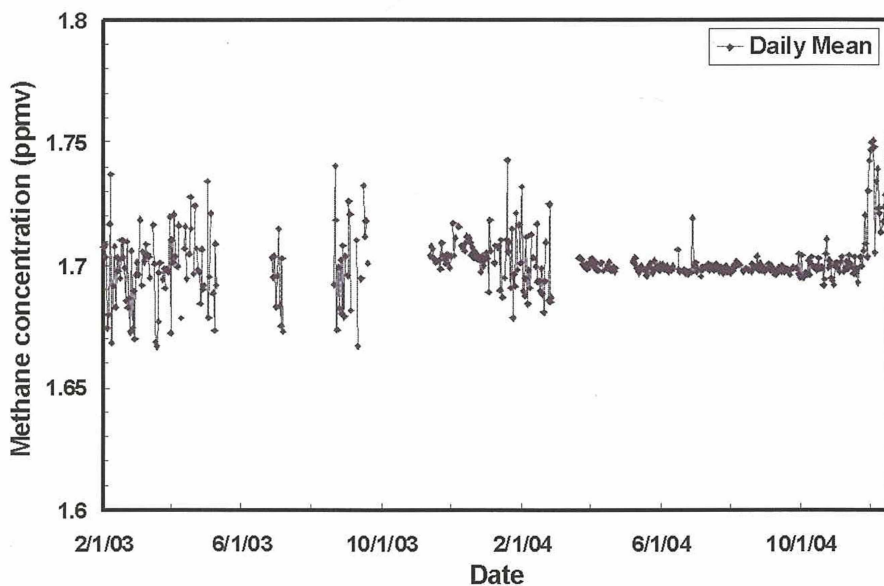


Fig. 4: Daily mean CH_4 concentration for the period 2003 and 2004

technical fault in the instrument, some times measurement could not be carried out at times. Measurements made over the period of 23 months showed day-to-day variation of a few ppb in surface CH₄ concentration at the site. As low as 1.67 ppmv and as high as 1.75 ppmv of daily mean CH₄ concentration has been observed. In general, hourly values were observed between 1.66 and 1.77 ppmv.

In order to study the growth of CH₄ in the atmosphere, daily mean values have been averaged to derive monthly mean CH₄ concentration, as depicted in Table 1.

Table 1 : Monthly mean CH₄ concentrations for the period 2003 and 2004 at Maitri

Month	No. of days	CH ₄ in ppmv	Standard deviation	No. of days	CH ₄ in ppmv	Standard deviation
	2003			2004		
January	0			26	1.704	0.014
February	28	1.695	0.016	23	1.696	0.012
March	26	1.697	0.014	11	1.701	0.002
April	22	1.704	0.013	19	1.700	0.002
May	8	1.699	0.021	25	1.699	0.002
June	3	1.701	0.005	23	1.699	0.005
July	7	1.692	0.015	28	1.698	0.001
August	11	1.697	0.020	25	1.699	0.002
September	11	1.704	0.020	30	1.698	0.002
October	0			30	1.699	0.004
November	15	1.703	0.003	27	1.700	0.013
December	24	1.706	0.005	16	1.721	0.015
Annual Average		1.699	0.013		1.701	0.006

Monthly mean CH₄ concentration has been observed to vary from 1.692 to 1.721 ppmv. Because of technical faults in the instrument, measurement could not be carried out in January 2003 and October 2003. As observed in Table 1, the mean yearly values for surface concentration of CH₄ during the observational period were nearly stable, exhibiting mean yearly value of 1.699 ppmv in 2003 with standard deviation of ± 0.013 and 1.701 ppmv in 2004 with standard deviation of ± 0.006 . Mean CH₄ concentration for the 22-month period was found to be 1.700 ppmv with standard deviation of ± 0.0095 . Similarly, no seasonal variation was observed at Maitri, Antarctica. The present data available for a period of 22 months does not show any evidence of increase in CH₄ concentration during that period. This is in agreement with observations by Dlugokencky *et al.*, (2003), who noted that the globally-averaged atmospheric methane abundance was constant at ~ 1751 ppb from 1999 through 2002, suggesting that during this four-year period the global methane budget has been at a steady state. However, our understanding is still not sufficient to examine whether the prolonged pause in CH₄ increase is temporary or permanent.

FREQUENCY DISTRIBUTION

To derive the nature of scatter in daily mean CH₄ concentration, frequency histogram against the concentration intervals; f_i (frequency interval) of 0.010 ppmv has been plotted as illustrated in Figure 5. Figure 5 (a) and (b) shows the frequency distribution for the period 2003 with $\Sigma f = 155$ and period 2004 with $\Sigma f = 281$ where $\Sigma f = N =$ Total number of measurements. Although the day-to-day variation of CH₄ has been observed from 1.67 to 1.75 ppmv, most values (about 83%) have been observed in the range from 1.685 to 1.715 ppmv. About 70% in 2003 and 90% in 2004 have been observed in the range from 1.685 to 1.715 ppmv. The exhibited small scatter of a few ppb in daily averages of near-surface atmospheric CH₄ throughout the observational period may be attributed to day-to-day variation in local meteorological conditions at the site.

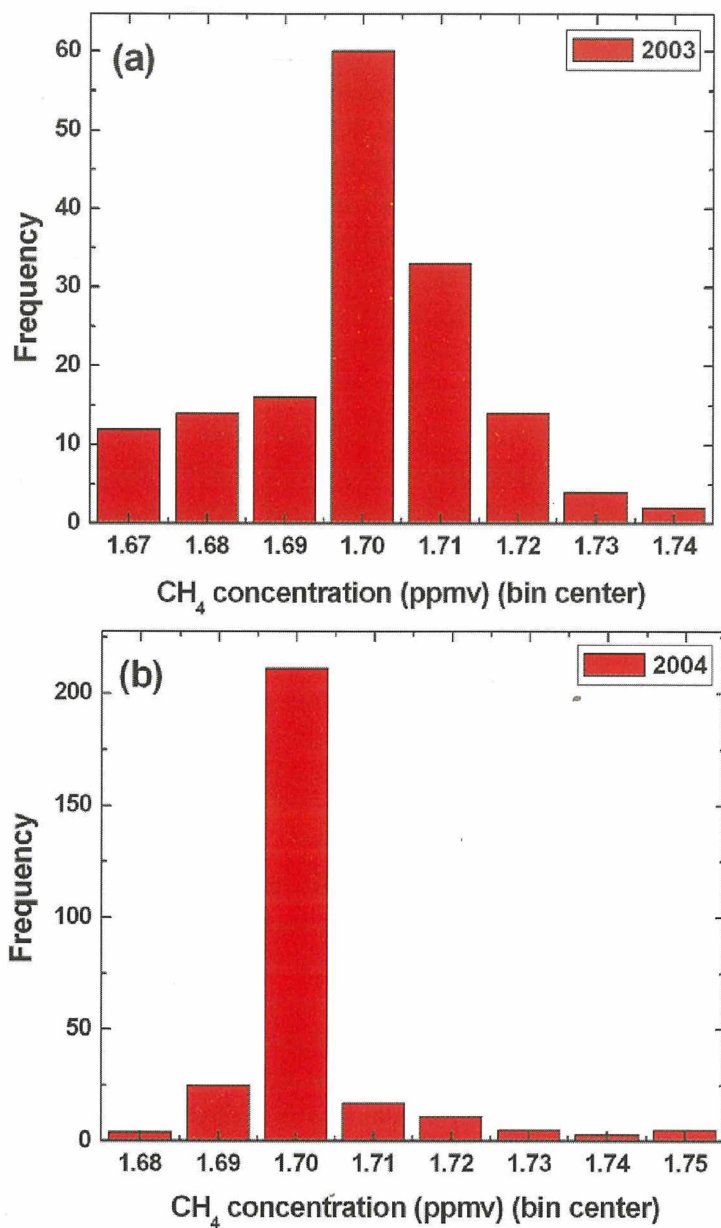


Fig. 5: Frequency distribution of measured CH₄ concentration at Maitri for the period (a) 2003 ($\Sigma f = 155$) and (b) 2004 ($\Sigma f = 281$) for f_i (frequency interval) = 0.010 ppmv ($\Sigma f = N =$ Total number of measurements).

II) CONCLUSIONS

The methane measurements carried out at the Indian Antarctic station, Maitri during 2003 and 2004 reveal that the atmospheric concentration of CH_4 is not increasing. Nearly invariant daily mean concentration of CH_4 observed at Maitri illustrates that the air at Maitri is well mixed and far removed from the source strength characteristics of land area. Hence is an ideal site for baseline measurements of trace gases in the atmosphere. The observed stable CH_4 concentration indicates a decrease in emissions, which may have forced the global methane budget towards a steady state. The reason for the change is still unclear. Dlugokencky et al (2003) believe that a major contributing factor was the break-up of the Soviet Union in 1991. Reduction in the oil and gas production as well as the industry became more efficient at plugging gas leaks from pipes and wells can also be one of the various reasons. They also noted that the methane emissions from the former Soviet Union dropped significantly throughout the 1990s due to lower fossil-fuel production. Based on current knowledge of the observed data, it is not possible to tell whether we are only observing a persistent but temporary pause in its increase or global levels of methane are likely to stabilize in the near future. However, to derive acceleration and deacceleration of global budget of these trace gases, it is necessary to have continuous measurements for a prolonged duration.

Most model simulations of Earth's climate indicate that an increase in the atmospheric concentration of a greenhouse gases will lead to an increase in the average surface air temperature of the Earth. CH_4 is the critical element in the greenhouse effect and second most important agent of potential future climate warming because of its large global warming potential, greenhouse forcing, its substantial projected future forcing and its long persistence in the atmosphere. To understand how the methane concentration will change global environment, it is necessary to have continuous measurement in a pollution free environment. Measurement of methane made for the period of 22 months showed that the day-to-day variation in CH_4 concentration was small except the variation of few ppb. No diurnal variation has been observed. About 83% of CH_4 values have been observed in the range of 1.685 ppmv to 1.715 ppmv during the entire period of measurement from February 2003 to December 2004.

The ambient air surface concentration of CH₄ during the period of observation was found to be nearly stable. Annual mean concentration of CH₄ has been observed at 1.699 ppmv in 2003 with standard deviation of 13 ppbv and 1.701 ppmv in 2004 with standard deviation of 6 ppbv. Mean CH₄ concentration for the 22 month period has been observed at 1.7 ppmv with standard deviation of 9 ppbv. The observations also illustrate that the baseline CH₄ concentration in the atmosphere is no more increasing. However, continuous measurement of methane at Maitri is required to understand the global methane dynamics, to reduce the uncertainties in the observations and to evaluate the effect of stable concentration of methane on its lifetime.

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